

Deep Reinforcement Learning from Human Preferences

Summary

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1 Problem Setup

Traditional reinforcement learning requires explicit reward functions to guide agent behavior. However, for complex tasks like robotics or game playing, designing good reward functions is difficult. This paper introduces a method to learn complex behaviors by collecting human preferences between trajectory segments, rather than explicit rewards. The goal is to train agents that perform well on tasks where reward functions are hard to specify but humans can recognize good outcomes when they see them.

2 Thesis

The central thesis is that human preference feedback can serve as a viable alternative to explicit reward functions in reinforcement learning. By training a reward model from human preferences and using this model to train the agent, we can achieve performance comparable to agents trained with hand-designed reward functions on complex tasks.

3 Key Concepts

Preference-Based Learning: Instead of receiving numerical rewards, the agent learns from human comparisons between trajectory segments. Humans simply indicate which of two short video clips shows better behavior.

Reward Model: A neural network that predicts which trajectory segment humans would prefer. This model learns to map trajectory segments to scalar reward values that align with human preferences.

Trajectory Segments: Short sequences of agent behavior (typically 25 timesteps or 1.7 seconds) that humans compare. These segments capture enough context to evaluate behavior while being short enough for humans to assess quickly.

Preference Elicitation: The process of collecting human feedback through a simple interface where humans select the better of two trajectory segments, with options for "tie" and "can't tell."

4 Methods

The approach consists of three main components:

1. **Data Collection:** Human contractors compare trajectory segments through a web interface. They receive specific instructions about what constitutes good behavior for each task (e.g., "move to the right as fast as possible" for robotics tasks).

2. **Reward Model Training:** A neural network (typically a CNN for visual tasks) is trained to predict which segment humans would prefer. The model takes trajectory segments as input and outputs reward predictions that are trained to match human preference data.
3. **Policy Optimization:** The reinforcement learning agent is trained using the learned reward model instead of hand-designed rewards. Standard RL algorithms like PPO can be used with the learned reward function.

The system continuously collects new preferences and retrains the reward model, allowing the agent to learn increasingly complex behaviors as human feedback accumulates.

5 Results

The authors demonstrate their method on several domains:

- **Atari Games:** The agent learns to play games like BeamRider, Breakout, and Pong using only human preferences, achieving performance comparable to agents trained with game scores.
- **Robotics Tasks:** The method works on MuJoCo robotics tasks like Hopper, Walker, and Ant, where the agent learns locomotion behaviors from human preferences about movement quality.
- **Sample Efficiency:** The approach requires relatively few human preferences (thousands rather than millions) to achieve good performance.

Key findings include that the reward model learns meaningful reward signals, the agent can generalize beyond the specific trajectories it was trained on, and human preferences can capture complex objectives that are difficult to encode in reward functions.

6 Significance

This work opens up new possibilities for reinforcement learning:

- **Complex Objectives:** Enables training agents for tasks with complex, multi-faceted goals that are hard to specify mathematically but easy for humans to recognize.
- **Reduced Reward Engineering:** Eliminates the need for careful reward function design, which has been a major bottleneck in applying RL to real-world problems.
- **Human-AI Collaboration:** Provides a natural way for humans to guide AI behavior without needing technical expertise in reward design.
- **Safety and Alignment:** Offers a path toward training AI systems that behave in ways humans actually want, rather than optimizing proxy metrics.

The paper established preference-based learning as a viable alternative to reward engineering and laid the foundation for subsequent work in reinforcement learning from human feedback (RLHF), which has become crucial for training large language models.